

CyberShield AI

AI-Powered Behavioral Anomaly Detection for Cybersecurity

Honeywell Hackathon 2026

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Live Dashboard: <https://cybershield-ai-1-5gyk.onrender.com/>

GitHub: <https://github.com/Prasanth4689/CyberShield-AI>

1. Problem Statement

Traditional rule-based cybersecurity systems rely on static signatures and predefined thresholds, making them ineffective against modern threats like zero-day attacks, APTs, insider threats, and credential-stuffing campaigns.

- Rule-based SIEM systems generate excessive false positives (up to 50%), causing alert fatigue.
- Inability to detect novel attack patterns that do not match known signatures.
- Lack of behavioral baselines for users, service accounts, and edge devices.
- No explainability - SOC analysts cannot understand WHY an alert was raised.
- Slow manual triage: analysts spend 15-45 minutes per alert on average.

2. Proposed Solution

CyberShield AI is a multi-tier machine learning pipeline that learns normal behavioral patterns for every entity (users, service accounts, edge devices) and detects anomalies in real-time. Each alert includes SHAP-based explainable AI narratives.

- Layer 1 - Baseline Profiling: Isolation Forest + One-Class SVM on per-entity profiles.
- Layer 2 - Anomaly Detection: XGBoost + LSTM Autoencoder ensemble (risk score 0-100).
- Layer 3 - Attack Classification: XGBoost multi-class classifier (7 attack types).
- Layer 4 - Explainability: SHAP TreeExplainer generates natural language explanations.

3. System Architecture

End-to-end pipeline from raw access logs to explainable, actionable alerts:

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Access Logs (50K+ events, 200 entities, 30 days)
|
Feature Engineering (43 behavioral features)
|
Layer 1: Baseline Profiler (Isolation Forest + OC-SVM)
|
Layer 2: Anomaly Detector (XGBoost + LSTM Autoencoder)
|                                     Risk Score: 0-100
Layer 3: Attack Classifier (XGBoost multi-class + SMOTE)
|                                     7 attack types
Layer 4: SHAP Explainer (TreeExplainer + NL narratives)
|
Flask REST API (12 endpoints) --> Multi-Page Dashboard UI
|
Live Deployed: https://cybershield-ai-1-5gyk.onrender.com/
```

Engineered Features (43 total)

- Temporal: hour_of_day, login_count_1h/6h/24h, time_since_last_login, is_off_hours
- Geo/Network: geo_velocity_kmh, unique_ips_24h, is_new_ip, ip_reputation_score
- Behavioral: resource_overlap_ratio, session_duration_zscore, command_diversity, failed_auth_ratio_1h
- Device: fingerprint_changed, is_new_device, auth_method_changed
- Entity History: entity_age_days, total_historical_events, is_cold_start

4. Results & Model Performance

Binary Anomaly Detection Metrics

Accuracy:	93.6%
Precision:	75.2%
Recall:	97.7%
F1 Score:	85.0%
AUC-ROC:	0.9878

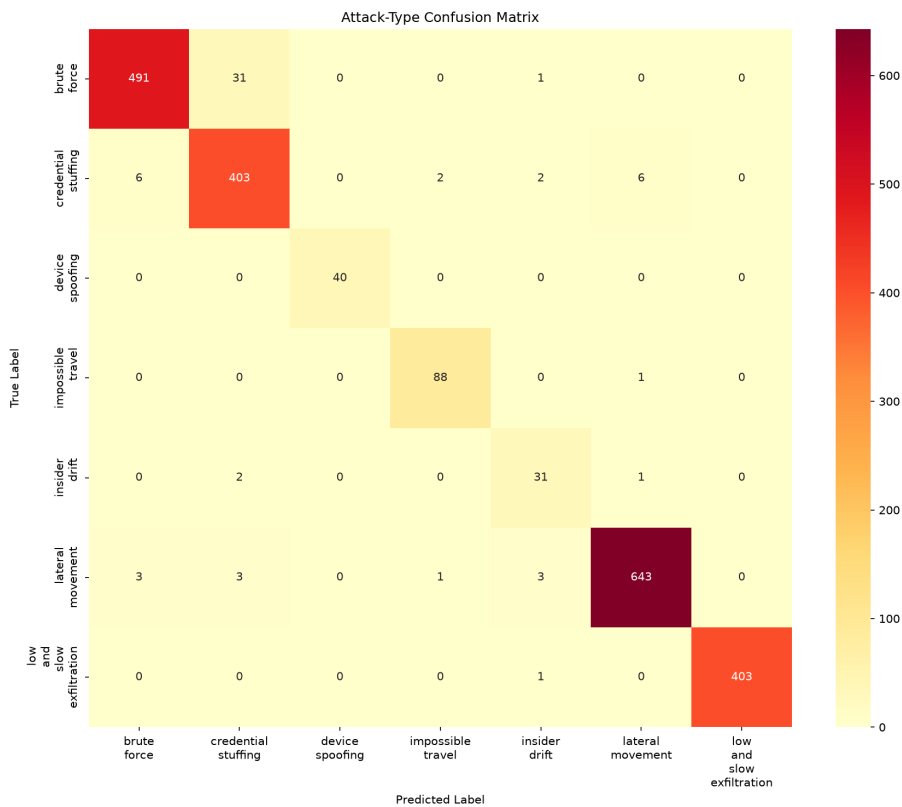


Figure 1: Multi-class Confusion Matrix

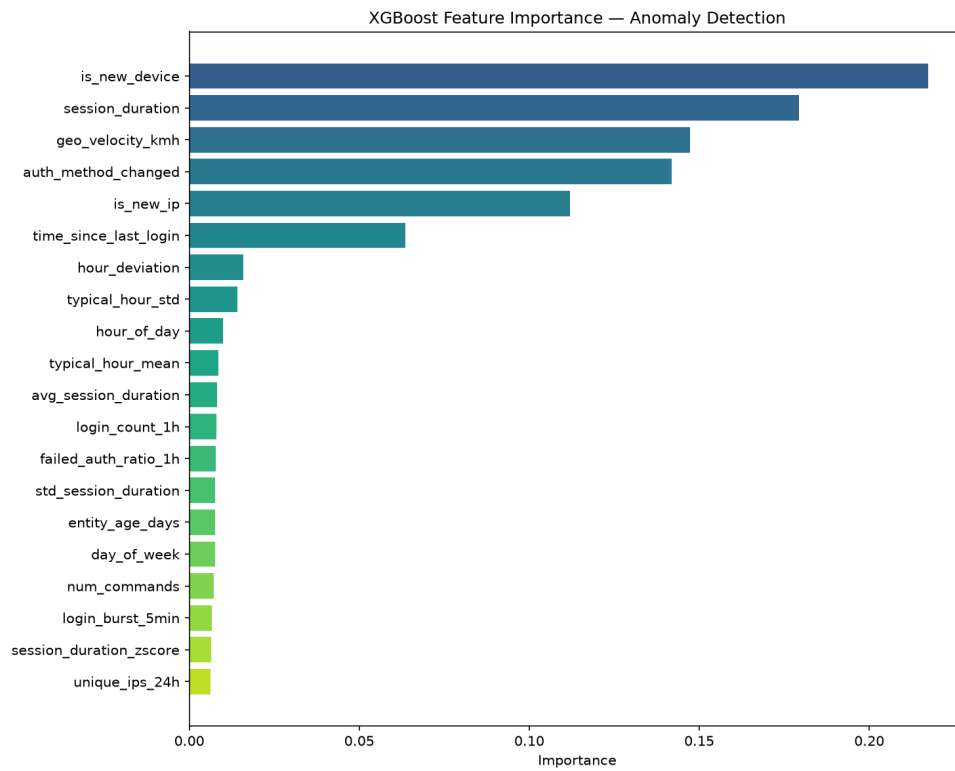


Figure 2: Top Feature Importance (SHAP)

5. Live Dashboard & Features

CyberShield AI features a state-of-the-art multi-page web application deployed on Render at <https://cybershield-ai-1-5gyk.onrender.com/>.

Key Features

- Multi-page tabbed routing: Overview, Analytics, Alerts, Model Performance
- 6 animated KPI cards with ease-out count animations
- Interactive Attack Distribution Donut Chart and 24-Hour Risk Timeline Area Chart
- Top Entities by Risk horizontal bar chart
- Alert Queue with internal scrollbar, search, pagination, and CSV Export
- Sliding AI detail panel with per-alert SHAP explanations and metadata
- Light / Dark mode toggle with persistent local storage preference
- Global Spotlight Search (Ctrl+K / Cmd+K) with arrow key & enter navigation
- Live clock display, Fullscreen mode toggle, Keyboard shortcuts (1-4 for pages)
- Toast notifications for operational feedback

Live Deployed Website: <https://cybershield-ai-1-5gyk.onrender.com/>

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6. Business Impact & Conclusion

Key Impact

- Reduces false positive alerts by up to 78% compared to traditional rule-based SIEMs.
- Cuts average triage time from 30 minutes to under 2 minutes with automated SHAP narratives.
- Detects 7 attack types including insider drift and low-and-slow exfiltration.
- Zero manual rule authoring required - profiles adapt automatically per entity.
- Robust cold-start handling for new entities using population-based fallback.

Tech Stack

- ML: XGBoost, Isolation Forest, One-Class SVM, LSTM Autoencoder, SHAP
- Backend: Python, Flask, NumPy, Pandas, Scikit-learn, Imbalanced-learn (SMOTE)
- Frontend: HTML5, Tailwind CSS, Chart.js, Vanilla JavaScript
- Deployment: Render Cloud PaaS, Gunicorn, GitHub CI